THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 1: **Exploring the Investigative World of Science** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student wants to find out how oil temperature affects the puffing of a puri. The factor she deliberately changes is called the: 1

(a) conclusion **(b)** dependent variable
(c) independent variable **(d)** constant

2. A scientific investigation usually begins with: 1

(a) a question or curiosity about an observation
(b) announcing the result
(c) buying equipment
(d) writing a final report

3. In an investigation of how the length of a pendulum affects its time period, the independent variable is: 1

(a) colour of the bob **(b)** length of the pendulum
(c) mass of the room **(d)** time period

4. In the same study, the time taken by the puri to puff up is the: 1

(a) dependent variable **(b)** independent variable
(c) hypothesis **(d)** control factor

5. The scientific method is best described as: 1

(a) a flexible approach that may be repeated and revised
(b) guessing the answer
(c) copying a friend's result
(d) a fixed list of steps always followed in one order

6. Two students get different answers for the same experiment. The most sensible step is to: 1

(a) argue until one gives up
(b) average without checking
(c) check the method and repeat the experiment
(d) ignore both results

7. Which of these must be kept the same in a fair test on puri puffing? 1

(a) the conclusion **(b)** the result obtained
(c) temperature being tested **(d)** thickness of the rolled dough

8. If the results do not support a hypothesis, a scientist should: 1

(a) change the data to fit **(b)** stop science
(c) hide the data **(d)** revise the hypothesis and test again

9. Which of the following shows a scientific attitude? 1

(a) refusing to test ideas **(b)** curiosity and willingness to question
(c) following rumours **(d)** believing without evidence

10. A tentative explanation that can be tested by an experiment is called a: 1

(a) measurement **(b)** law **(c)** hypothesis **(d)** theory only

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What is a hypothesis? Give one example related to puri making. 2

12. Why do scientists repeat an experiment several times? 2

13. State two safety rules to be followed while working with hot oil or a flame. 2

14. What is a variable? Name the two kinds of variables used in an investigation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe how you would plan an investigation to find the effect of oil temperature on the time taken by a puri to puff up. 3

16. Why is the scientific method not a rigid list of steps? Explain with a situation. 3

17. What is meant by a fair test? Why does a conclusion from an unfair test cause confusion? 3

OR

List three factors that must be kept constant while testing the effect of the thickness of dough on the puffing of a puri. 3

18. Give three qualities of a good investigator. 3

OR

Why is it important to take measurements using standard units and instruments? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Design a complete investigation to study whether the amount of water used in kneading dough affects how a puri puffs. Mention question, hypothesis, variables, constants, procedure and recording. 5

OR

Describe how errors can occur while taking measurements and how they can be reduced. 5

20. Explain the steps of the scientific method using the puri-puffing investigation. Why is it called a cycle? 5

OR

Write the safety precautions needed when a student fries puris or heats liquids in the laboratory. 5

END OF QUESTION PAPER — Total: 40 Marks